

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

PARKING AID CONTROL MODULE (PACM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Parking Aid Control Module (PACM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Parking Aid](#) (413-13 Parking Aid,

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B129C-12	Left Front Side Sensor - Circuit short to battery	- Front left side park assist sensor signal circuit short circuit to power - Front left side park assist sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B129C-14	Left Front Side Sensor - Circuit short to ground or open	- Front left side park assist sensor signal circuit short circuit to ground, open circuit, high resistance - Front left side park assist sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new

			position of the suspect sensor, install a new sensor
B129C-96	Left Front Side Sensor - Component internal failure	- Front left side park assist sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left side park assist sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit - Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B129D-12	Right Front Side Sensor - Circuit short to battery	- Front right side park assist sensor signal circuit short circuit to power - Front right side park assist sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the

			suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B129D-14	Right Front Side Sensor - Circuit short to ground or open	- Front right side park assist sensor signal circuit short circuit to ground, open circuit, high resistance - Front right side park assist sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B129D-96	Right Front Side Sensor - Component internal failure	- Front right side park assist sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front right side park assist sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit - Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and

			damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B129E-73	Enable/Disable Switch - Semi-automatic Advanced Parallel Park - Actuator stuck closed	- Park assist switch signal circuit short circuit to ground - Park assist switch stuck active - Park assist switch failure	- Check the operation of the park assist switch - Refer to the electrical circuit diagrams and check the park assist switch signal circuit for short circuit to ground
B129F-11	Function LED - Semi-Automatic Advanced Parallel Park - Circuit short to ground	- Park assist switch circuit short circuit to ground - Park assist switch failure	- Refer to electrical wiring diagrams and check the parallel park LED circuit for short circuit to ground. Repair or install a new wiring harness as necessary - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202). If the problem persists, suspect the switch/LED
B129F-12	Function LED - Semi-Automatic Advanced Parallel Park - Circuit short to battery	- Park assist switch circuit short circuit to power - Park assist switch failure	- Refer to electrical wiring diagrams and check the parallel park LED circuit for short circuit to power. Repair or install a new wiring harness as necessary - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self

			Test (0x0202). If the problem persists, suspect the switch/LED
B129F-13	Function LED - Semi-Automatic Advanced Parallel Park - Circuit open	▪ Park assist switch circuit open circuit, high resistance ▪ Park assist switch failure	▪ Refer to electrical wiring diagrams and check the parallel park LED circuit for open circuit, high resistance. Repair or install a new wiring harness as necessary ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202). If the problem persists, suspect the switch/LED
B1443-12	Left Rear Side Proximity Sensor - Circuit short to battery	▪ Rear left side park assist sensor signal circuit short circuit to power ▪ Rear left side park assist sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1443-14	Left Rear Side Proximity Sensor - Circuit short to ground or open	▪ Rear left side park assist sensor signal circuit short circuit to ground, open circuit, high resistance ▪ Rear left side park assist sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved

			diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1443-96	Left Rear Side Proximity Sensor - Component internal failure	▪ Rear left side park assist sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left side park assist sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit ▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1454-12	Right Rear Side Proximity Sensor - Circuit	▪ Rear right side park assist sensor signal	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage. Check sensor circuit for

	short to battery	circuit short circuit to power ▪ Rear right side park assist sensor failure	short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1454-14	Right Rear Side Proximity Sensor - Circuit short to ground or open	▪ Rear right side park assist sensor signal circuit short circuit to ground, open circuit, high resistance ▪ Rear right side park assist sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1454-96	Right Rear Side Proximity Sensor - Component	▪ Rear right side park assist sensor signal circuit short	

		internal failure circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right side park assist sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit ▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B36-12	Right Front Outer Sensor - Circuit short to battery	▪ Front right outer parking aid sensor signal circuit short circuit to power ▪ Front right outer parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the front bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free

			sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B36-14	Right Front Outer Sensor - Circuit short to ground or open	- Front right outer parking aid sensor signal circuit short circuit to ground, open circuit, high resistance - Front right outer parking aid sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B36-96	Right Front Outer Sensor - Component internal failure	- Front right outer parking aid sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front right outer parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit - Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved

			diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B38-12	Right Front Inner Sensor - Circuit short to battery	▪ Front right inner parking aid sensor signal circuit short circuit to power ▪ Front right inner parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the front bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B38-14	Right Front Inner Sensor - Circuit short to ground or open	▪ Front right inner parking aid sensor signal circuit short circuit to ground, open circuit, high resistance ▪ Front right inner parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and

			perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B38-96	Right Front Inner Sensor - Component internal failure	▪ Front right inner parking aid sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front right inner parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit ▪ Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B40-12	Left Front Outer Sensor - Circuit short to battery	▪ Front left outer parking aid sensor signal circuit short	▪ Refer to electrical wiring diagrams and check the front bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install

		circuit to power Front left outer parking aid sensor failure	a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B40-14	Left Front Outer Sensor - Circuit short to ground or open	- Front left outer parking aid sensor signal circuit short circuit to ground, open circuit, high resistance - Front left outer parking aid sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B40-96	Left Front Outer Sensor - Component internal failure	Front left outer parking aid sensor signal circuit short circuit to	

		ground, short circuit to power, open circuit, high resistance Front left outer parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit - Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B42-12	Left Front Inner Sensor - Circuit short to battery	- Front left inner parking aid sensor signal circuit short circuit to power - Front left inner parking aid sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free

			sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B42-14	Left Front Inner Sensor - Circuit short to ground or open	- Front left inner parking aid sensor signal circuit short circuit to ground, open circuit, high resistance - Front left inner parking aid sensor failure	- Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) - If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B42-96	Left Front Inner Sensor - Component internal failure	- Front left inner parking aid sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left inner parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit - Refer to electrical wiring diagrams and check the front bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary - Check the connector for integrity and damage, then re-connect sensor to confirm connection - Using the manufacturer approved

			diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B44-12	Right Rear Outer Sensor - Circuit short to battery	▪ Rear right outer parking aid sensor signal circuit short circuit to power ▪ Rear right outer parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B44-14	Right Rear Outer Sensor - Circuit short to ground or open	▪ Rear right outer parking aid sensor signal circuit short circuit to ground, open circuit, high resistance ▪ Rear right outer parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and

			perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B44-96	Right Rear Outer Sensor - Component internal failure	▪ Rear right outer parking aid sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right outer parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit ▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, short circuit to power, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B46-12	Right Rear Inner Sensor - Circuit short to	▪ Rear right inner parking aid sensor signal	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage. Check sensor circuit for

	battery	circuit short circuit to power ▪ Rear right inner parking aid sensor failure	short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B46-14	Right Rear Inner Sensor - Circuit short to ground or open	▪ Rear right inner parking aid sensor signal circuit short circuit to ground, open circuit, high resistance ▪ Rear right inner parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B46-96	Right Rear Inner Sensor - Component internal failure	▪ Rear right inner parking aid sensor signal circuit short	

		circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right inner parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit ▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, short circuit to power, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B48-12	Left Rear Outer Sensor - Circuit short to battery	▪ Rear left outer parking aid sensor signal circuit short circuit to power ▪ Rear left outer parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange

			the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B48-14	Left Rear Outer Sensor - Circuit short to ground or open	▪ Rear left outer parking aid sensor signal circuit short circuit to ground, open circuit, high resistance ▪ Rear left outer parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B48-96	Left Rear Outer Sensor - Component internal failure	▪ Rear left outer parking aid sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left outer parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit ▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, short circuit to power, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection

			▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B50-12	Left Rear Inner Sensor - Circuit short to battery	▪ Rear left inner parking aid sensor signal circuit short circuit to power ▪ Rear left inner parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage. Check sensor circuit for short circuit to power. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B50-14	Left Rear Inner Sensor - Circuit short to ground or open	▪ Rear left inner parking aid sensor signal circuit short circuit to ground, open circuit, high resistance ▪ Rear left inner parking aid sensor failure	▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved

			diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B50-96	Left Rear Inner Sensor - Component internal failure	▪ Rear left inner parking aid sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left inner parking aid sensor failure	NOTE: This DTC will log if the local ground of the sensor is open circuit ▪ Refer to electrical wiring diagrams and check the rear bumper harness for damage/corrosion. Check sensor circuit for short circuit to ground, short circuit to power, open circuit. Repair or install a new wiring harness as necessary ▪ Check the connector for integrity and damage, then re-connect sensor to confirm connection ▪ Using the manufacturer approved diagnostic system clear the DTCs and perform routine - On Demand Self Test (0x0202) ▪ If the problem persists remove the suspect sensor from the bumper. Inspect the sensor connector for signs of water ingress/corrosion. Exchange the relevant sensor with a fault-free sensor. Clear the DTCs and perform routine - On Demand Self Test (0x0202). If a DTC is set for the new position of the suspect sensor, install a new sensor
B1B54-11	Function LED - Park Aid -	▪ Parking aid switch LED	▪ Refer to the electrical circuit diagrams and check the parking aid switch LED

	Circuit short to ground	circuit short circuit to ground	circuit for short circuit to ground
B1B54-12	Function LED - Park Aid - Circuit short to battery	▪ Parking aid switch LED circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the parking aid switch LED circuit for short circuit to power
B1B54-13	Function LED - Park Aid - Circuit open	▪ Parking aid switch LED circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the parking aid switch LED circuit for open circuit, high resistance
B1B57-11	Front Sensors Power Circuit - Circuit short to ground	▪ Front park assist / parking aid sensors power circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the front park assist / parking aid sensors power circuit for short circuit to ground
B1B57-12	Front Sensors Power Circuit - Circuit short to battery	▪ Front park assist / parking aid sensors power circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the front park assist / parking aid sensors power circuit for short circuit to power
B1B57-15	Front Sensors Power Circuit - Circuit short to battery or open	▪ Front park assist / parking aid sensors power circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the front park assist / parking aid sensors power circuit for short circuit to power, open circuit, high resistance
B1B58-11	Rear Sensors Power Circuit - Circuit short to ground	▪ Rear park assist / parking aid sensors power circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear park assist / parking aid sensors power circuit for short circuit to ground

B1B58-12	Rear Sensors Power Circuit - Circuit short to battery	▪ Rear park assist / parking aid sensors power circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear park assist / parking aid sensors power circuit for short circuit to power
B1B58-15	Rear Sensors Power Circuit - Circuit short to battery or open	▪ Rear park assist / parking aid sensors power circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear park assist / parking aid sensors power circuit for short circuit to power, open circuit, high resistance
B1C30-73	Disable Switch - Park Aid - Actuator stuck closed	▪ Parking aid switch signal circuit short circuit to ground ▪ Parking aid switch stuck active ▪ Parking aid switch failure	▪ Check the operation of the parking aid switch ▪ Refer to the electrical circuit diagrams and check the parking aid switch signal circuit for short circuit to ground
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN chassis systems bus	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive / sequence counter incorrect / not updated	▪ Invalid data received from another control module via the high speed CAN chassis systems bus	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-83	High Speed CAN Communication Bus - Value of	▪ Invalid data received from another control module via the	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module.

	signal protection calculation incorrect	high speed CAN chassis systems bus	Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	Missing message from another control module via the high speed CAN chassis systems bus	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN chassis systems bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN chassis systems bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-57	Internal Control Module Software Incompatibility - Invalid / incomplete software component	- Car configuration file mismatch with vehicle specification - Incorrect parking aid control module installed	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs - Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest - Install a new parking aid control module as necessary
U100B-00	Semi-Automatic Parallel Parking System - No sub type information	- Power steering control module power or ground circuit open circuit, high resistance - High speed CAN chassis	- Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check

		systems bus circuit short circuit to ground, short circuit to power, open circuit, high resistance Power steering system fault	the high speed CAN chassis systems bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index
U2300-54	Central Configuration - Missing calibration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U2300-55	Central Configuration - Not configured	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U2300-56	Central Configuration - Invalid / incomplete configuration	Car configuration file mismatch with vehicle specification	

			NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U2300-64	Central Configuration - Signal plausibility failure	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U3000-47	Control Module - Watchdog / safety microcontroller failure	Parking aid control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new parking aid control module
U3000-49	Control Module - Internal electronic failure	Parking aid control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new parking aid control module
U3000-55	Control Module - Not configured	Parking aid control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the parking aid control module with the latest level software

U3000-56	Control Module - Invalid / incomplete configuration	▪ Parking aid control module is not configured correctly	▪ Using the manufacturer approved diagnostic system, re-configure the parking aid control module with the latest level software
U3000-63	Control Module - Circuit / component protection time-out	▪ Parking aid control module internal circuit protection routine has deactivated the module following detection of repeated parking aid sensor short circuit faults	NOTE: This DTC is only logged if the control module has detected persistent parking aid sensor short circuit faults. Ensure that all circuit faults have been resolved before proceeding with the reset short protection routine ▪ Check for related parking aid / park assist sensor circuit fault DTCs and ensure all parking aid / park assist sensor faults are resolved ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Reset/Clear Specified Function (Reset Short Protection) (0x040E) - to re-enable the parking aid control module output circuits